

Design and Implementation of an Eye Blinking Detector System for Automobile Accident Prevention

Tariq Jamil, Iftaquaruddin Mohammed, and Medhat H. Awadalla

Department of Electrical and Computer Engineering

Sultan Qaboos University

AlKhod, Muscat, OMAN

tjamil@squ.edu.om

Abstract—Ever increasing number of fatal traffic accidents around the world can be significantly reduced if modern technology is incorporated within the automobile to assess the physical condition of the driver at regular intervals during the movement of the vehicle and preventive measures are automatically taken for the safety of all concerned entities, both within the vehicle and outside the vehicle. In this paper, design of an eye blinking detector system is presented which can monitor the physical state of the driver at regular intervals during his/her driving and, if needed, can raise an audible alarm within the vehicle to alert the driver or initiate application of vehicle's braking system. In case of multiple failures to raise the alertness level of the driver, the designed system can automatically inform the law enforcement authorities about the rogue driver on the road. The prototype of the design has been successfully implemented which leads to the conclusion that such a system can help in keeping the driver awake at all times while driving and hence facilitate in avoidance of any traffic accidents involving driver's alertness.

Keywords—eye blinking, accident avoidance, driving, alertness, vehicle safety.

I. INTRODUCTION

One of the main reasons attributed to the increasing number of traffic accidents around the world is the drowsiness of the driver which results in either complete or partial impairment to control the vehicle while in motion, and therefore puts the life of the driver and others in danger. Traffic police, in its effort to get rid of drunk drivers, administers breathalyser test for those drivers who are on the road but there is currently no mechanism to monitor the driver's condition while he is driving the vehicle.

Driving Under Influence (DUI) can cause vehicle accidents which leads to disability and death in worst cases. Additionally, drowsiness of the drivers resulting from taking sedative medicines irresponsibly or tiredness and fatigue of the drivers also contribute to road accidents. It is therefore crucial to develop a monitoring system within the vehicle which can keep an eye on the health of the driver while driving and can alert firstly the driver and, in case of nonresponsive driver, the law enforcement authorities about the condition of the driver and presence of a possible accident-prone vehicle on the road. This can result in reducing the number of accidents and can save precious lives and assets. [1]

There is no known device currently installed in the vehicle which can determine whether the driver is drowsy or not. In this paper, we have proposed and implemented a system which can identify whether the driver is drowsy or not by monitoring the blinking pattern of his/her eyes and then raise an audible alarm in the vehicle in an effort to alert the driver. If the driver does not respond positively within a certain time interval, the system has the capability of applying brakes to the vehicle and alerting law enforcement authorities about the rogue driver on the road. We have limited ourselves to the discussion of only the electrical components and working of designed eye blinking system in this paper, and have left out the details of mechanical braking system and mechanism of communication with the law enforcement authorities as supplementary research work for the future.

II. PREVIOUS WORK

There have been several designs proposed in scientific literature to detect driver's drowsiness and subsequently to control the vehicle. A warning system to detect driver's drowsiness by applying image processing techniques on images captured through a video camera and then conducting analysis utilizing a neural network has been proposed by Singh and Banga [2]. The first step in this design is the image acquisition which is done by using video camera installed on the dashboard and then converts it into image frames. The second step is the face detection using Viola Jones Algorithm, followed by the extraction of the eyes' features. Lastly, a neural network trained using back-propagation algorithm on images of eyes in various states (open, drowsy, closed) is used to predict whether the driver is drowsy or not. An accuracy of 96.7% was obtained by the authors in correctly assessing the drowsiness state of the driver using this technique. Another driver-fatigue monitoring system which uses remotely-located charge-coupled device (CCD) cameras equipped with active infra-red illuminators to take video images of the drivers has been designed by Varma et. al. [3]. Various visual cues that typically characterize the level of alertness of a person, such as eyelid movement, gaze movement, head movement, and facial expression, are extracted in real time and systematically combined in a probabilistic model to infer the fatigue level of the driver. This results in a much more robust and accurate fatigue characterization than using a single visual cue.

Dharani and Lalitha [4] have designed a non-intrusive system to determine driver's fatigue using infra-red sensor to detect eye blinking and, in case of confirmed drowsiness, warns the driver with an alarm and then automatically controls and stops the vehicle. Turan and Gupta [5] have proposed a system to detect driver's drowsiness by measuring the changes in the geometric features of the driver's face (eyes and mouth) during yawning. Morris et al. [6] have proposed the eye blinking detection method using spatio-temporal filtering and Lucas-Kanade feature tracking to locate the position of the head and to extract eyelid movements. They have reported 95% eye-blink detection result but head movements affect the variance map computation and cause a sharp drop in performance. Sirohey et al. [7] have proposed an approach for determining eye blinks by locating eye corners, eyelids and irises in each frame of an image sequence, and analyzing their movements to determine changes in gaze direction and blinking respectively. The authors claim that their algorithm can track iris and eyelid motion more than 90% of the time, but not in real time. Similar works include Grauman et al. [8], Chau et al. [9], Heishman et al. [10], Pan et al. [11], Orozco et al. [12], Bacivarov et al. [13], Divijack et al. [14], Lee et al. [15], and Naveed et al. [16], each reporting over 90% accuracy in detecting eye blinking using various techniques.

III. DESIGN

A. Motivation

Oman lies between latitudes 16° and 28° N, and longitudes 52° and 60° E, and has a vast expanse of area, 309,501 sq. km (approximately 119,498 sq. miles), covered mostly with desert plain in the center, and mountain ranges along the north (Al Hajar Mountains) and southeast coast. The country's main cities are located along the coast: the capital city Muscat, Sohar and Sur in the north, and Salalah in the south. Oman's climate is hot and dry in the interior and humid along the coast [17]. In 2015, the population of Oman is estimated to be 4,296,168 and it is ranked 125/198 worldwide in terms of number of people resident in Oman [18]. Despite having relatively small population, Oman has one of the highest numbers of road accident deaths (35 per 100,000) compared to the global average of 19 per 100,000 and 5.4 per 100,000 for United Kingdom [19]. This translates to almost 1 death per 4000 of inhabitants of Oman, almost double the global average, and 6 times to that of United Kingdom and the developed world. Major reasons for this high number of road accidents include coming of age of an overwhelmingly young population for whom driving fast and recklessly is a passion (64% under 21 years) commonly fueled by Hollywood blockbusters such as "Fast and Furious", fatigue due to driving long distances without any suitable rest-stops along the highways, presence of medicinal or recreational drugs, alcohol abuse, poor eyesight, poor vehicle maintenance, defects in road design, and poor management of road construction activity.

Keeping in mind the alarming statistics of road accident deaths in Oman and with the aim of contributing towards the

reduction in these figures, we decided to design an affordable and simple eye-blinking detection system which could be easily installed by the driver in his car and alert the driver in case he falls asleep while driving, or, as last resort, alert the law enforcement authorities of a rogue driver on the road.

B. Hardware

The main components of our system are shown in Fig. 1: (i) Microcontroller (ii) Eye Blink Detector (USB camera) (iii) LCD display (iv) Power supply, and (v) Motor (speed control module).

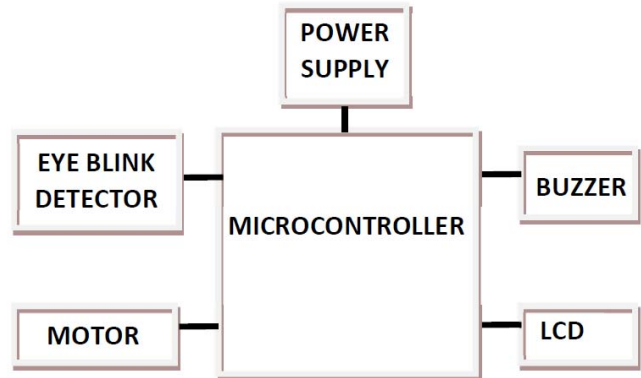


Fig. 1. Block Diagram of Eye-Blinking Detection System

Microcontroller is the heart of our system which is responsible for synchronizing among all sub-components, and also plays the role of the decision maker when to sound the buzzer, instruct the motor to apply the vehicle braking system, or to contact law enforcement authorities in case the first two options do not heed desired results. The microcontroller is powered by DC supply, commonly available from the cigarette-lighter knob in most car's dashboards. The images captured by the eye blink detector (USB camera) are fed into the microcontroller which uses Haar cascade classifier algorithm to detect both left and right eyes in the images and their current states [20]. If the current states for either one or both eyes are detected to be closed for at least a few seconds, the LCD displays a warning message and the buzzer starts sounding. The microcontroller can be interfaced to the vehicle braking system which can start applying the brakes slowly if warranted by the condition of the driver. A networking card can also be connected with the microcontroller to send a message to the law enforcement authorities, if desired.

C. Software

In our design, we have used OpenCV for real-time eye blinking detection and tracking. OpenCV is a library of programming functions focusing on real time image processing, and blinking of the eyes is tracked by estimating eye-position within high or low occlusion conditions using centroid analysis technique [21]. This technique has been shown to give an average success rate of over 96% in correctly detecting and tracking blinking eyes under average to good lighting conditions.

D. How does the system work?

Our system starts functioning as soon as the driver turns the key to start the car's engine. The USB Camera, installed at a suitable position on the dashboard is used to monitor the state of driver's eyes while the car is in motion. The camera transmits the real time eyes tracking information to the microcontroller installed in our system. After the microcontroller receives the information coming from the camera, it starts to compare the difference between the successive readings of the eye tracking information. If both the eyes are open, the system will do nothing. However, if any one or both the eyes are detected to remain closed for a few seconds, the system activates the buzzer and the driver has a limited amount of time to respond and turn it off. In case of repeated buzzer sounds, the vehicle braking system gets activated through the motor. As a last step, the law enforcement authorities are notified about the rogue driver on the road.

IV. IMPLEMENTATION

The designed system was successfully implemented using Arduino UNO Board (containing ATmega 328P microcontroller) [22], LCD03 – I2C/Serial LCD [23], HYT-3015A Buzzer [24], an ordinary USB camera, and a few simple resistances. The total cost of the components was only US\$48 which is quite affordable.

V. CONCLUSION

Several studies indicate that drowsiness is one of the most fatal causes of vehicle accidents around the world. Thus it is crucial to study this issue and provide solutions to prevent such accidents. This paper has presented a real-time eye-blinking detection system which can be easily installed by the driver on the dashboard and can monitor the status of driver's alertness. During the course of this project, we also compared USB camera technology with infra-red sensor technology for detecting eye movement, and decided on using USB camera mainly because of its ubiquitous availability and ease of operation. On the other hand, IR sensor provided better eye blinking detection success rate in night-time environment compared to USB camera.

VI. ACKNOWLEDGMENTS

The project was completed in the Department of Electrical and Computer Engineering at Sultan Qaboos University (Oman) as a senior-year project by a group of three students, Nassr Al-Salmi, Haitham Al-Mahrooqi, and Said Al-Mahrooqi, under the supervision of the authors.

REFERENCES

- [1] H. Al-Qareeni, "Traffic safety institute systematic knowledge to enhance road safety," Royal Oman Police Articles. Available from URL: <http://www.rop.gov.om/english/articledetails.asp?articleid=24> (Last accessed on 1-Nov-2015).
- [2] I. Singh and V.P. Banga, "Development of a drowsiness warning system using neural network," International Journal of Advanced Research in Electrical, Electronics, and Instrumentation Engineering, Vol. 2, No. 8, August 2013, pp. 3614-3623.
- [3] A. R. Varma, S. V. Arote, C. Bharti, and K. Singh, "Accident prevention using eye blinking and head movement," IJCA Proceedings on Emerging Trends in Computer Science and Information Technology, April 2012, ETCSIT(4), pp. 31-35.
- [4] M. Dharani and B. Lalitha, "Automatic vehicle control using sensors," International Journal of Innovative Research and Development, Vol. 2, No. 3, March 2013, pp. 744-750.
- [5] G. Turan and S. Gupta, "Road accidents prevention system using driver's drowsiness detection", International Journal of Advanced Research in Computer Engineering and Technology, Vol. 2, No. 11, November 2013, pp. 2981-2983.
- [6] T. Morris, P. Blenkhorn, and F. Zaidi, "Blink detection for real-time eye tracking," Journal of Network and Computer Applications, Vol.25, No. 22, 2002, pp. 129-143.
- [7] S. Sirohey, A. Rosenfeld, and Z. Duric, "A method of detecting and tracking irises and eyelids in video," Pattern Recognition, Vol. 35, No. 6, 2002, pp. 1389-1401.
- [8] K. Grauman, M. Betke, J. Lombardi, J. Gips, G. R. Bradski, "Communication via eye blinks and eyebrow raises: Video-based human-computer interfaces," 2003, Springer-Verlag, pp. 359-373.
- [9] M. Chau and M. Betke. 2005. Real Time Eye Tracking and Blink Detection With USB Cameras. Technical Report, Boston University Computer Science.
- [10] R. Heishman, and Z. Doric. 2007. Using Image Flow to Detect Eye Blinks in Color Video. In Proceedings of the IEEE Workshop on Applications of Computer Vision, pp. 52.
- [11] G. Pan, L. Sun, Z.Wu, and S. Lao. 2007. Eye BlinkBased Anti-Spoofing in Face Recognition from A Generic Web Camera. In Proceedings of the 11th IEEE international conference on computer vision, pp. 1-8.
- [12] J. Orozco, F. Xavier Roca, and J. González. 2009. RealTime Gaze Tracking with Appearance-Based Models. Journal of Machine Vision and Applications, Vol.20, pp. 353-364.
- [13] I. Bacivarov, M. Ionita, and P. Corcoran. 2008. Statistical Models of Appearance for Eye Tracking and Eye-Blink Detection and Measurement. IEEE Transactions on Consumer Electronics, Vol.54, No.3, pp. 1312-1320.
- [14] M. Divijak, and H. Bischof. 2009. Eye Blink Based Fatigue Detection for Prevention of Computer Vision Syndrome. In Proceedings of the IAPR conference on machine vision Applications, pp. 10-14.
- [15] W. Lee, E. Lee, and K. Park. 2010. Blink Detection Robust to Various Facial Poses. Journal of Neuroscience Methods , pp. 356-372.
- [16] S. Naveed, B. Sikander, and M. Khiya. 2012. Eye Tracking System with Blink Detection. Journal of Computing, Vol.4, No.3.
- [17] <https://en.wikipedia.org/wiki/Oman> (Last accessed on 24 November 2015).
- [18] <http://www.ncsi.gov.om/Pages/NCSE.aspx> (Last accessed: 24 November 2015).
- [19] <http://www.salimandsalimah.org/oman-crashes/crash-statistics/> (Last accessed 24 November 2015).
- [20] D. Pimlaskar, M. Nagmode, and A. Borkar. 2013. Real Time Eye Blinking Detection and Tracking using OpenCV. International Journal of Engineering Research and Applications, Vo. 3, No. 5, pp. 1780-1787.
- [21] P. Viola and M. Jones. 2001. Rapid Object Detection using a boosted cascade of simple features. In Proceedings of the IEEE International Conference and Computer Vision and Pattern Recognition, Vol. 1, pp. 511-518.
- [22] <https://www.arduino.cc/en/main/arduinoBoardUno> (Last accessed on 30 November 2015).
- [23] <http://www.robot-electronics.co.uk/htm/Lcd03tech.htm> (Last accessed on 30 November 2015).
- [24] http://emilychen2823.diytrade.com/sdp/859069/4/pd-4441735/5251868-1752642/piezoelectric_buzzer_HYT-3015A.html (Last accessed on 30 November 2015).